COMPUTER NETWORKS – PRACTICAL JOURNAL

RIP Routing and VLAN Configuration in Cisco Packet Tracer

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Aim

To configure dynamic routing using RIP between two routers, and then to create VLANs on a switch to separate departments (Purchase and Sales), and verify connectivity.

Requirements

Cisco Packet Tracer; 2 × Router 1841; 2 × Switch 2950-24; 3 × PC; 2 × Laptop; copper straight-through cables (router-switch, switch-PC) and copper cross-over cable (router-router).

IP Addressing Table

Device / Interface	IP Address	Subnet Mask	Default Gateway
PC0 – Fa0	192.168.1.2	255.255.255.0	192.168.1.1
PC1 – Fa0	192.168.2.2	255.255.255.0	192.168.2.1
PC2 – Fa0	192.168.2.3	255.255.255.0	192.168.2.1
Laptop0 – Fa0	192.168.2.4	255.255.255.0	192.168.2.1
Laptop1 – Fa0	192.168.2.5	255.255.255.0	192.168.2.1
Router0 – Fa0/0	10.0.0.1	255.0.0.0	—
Router0 – Fa0/1	192.168.1.1	255.255.255.0	—
Router1 – Fa0/0	10.0.0.2	255.0.0.0	—
Router1 – Fa0/1	192.168.2.1	255.255.255.0	—

PART A: RIP Configuration

A1. Build the Topology

Step 1: Place Router0, Router1, Switch0, Switch1, PC0, PC1 and PC2 and connect them as shown. Router0 Fa0/0 to Router1 Fa0/0 (cross-over); Router0 Fa0/1 to Switch0 Fa0/1; Switch0 Fa0/2 to PC0; Router1 Fa0/1 to Switch1 Fa0/1; Switch1 Fa0/2 to PC1; Switch1 Fa0/3 to PC2.

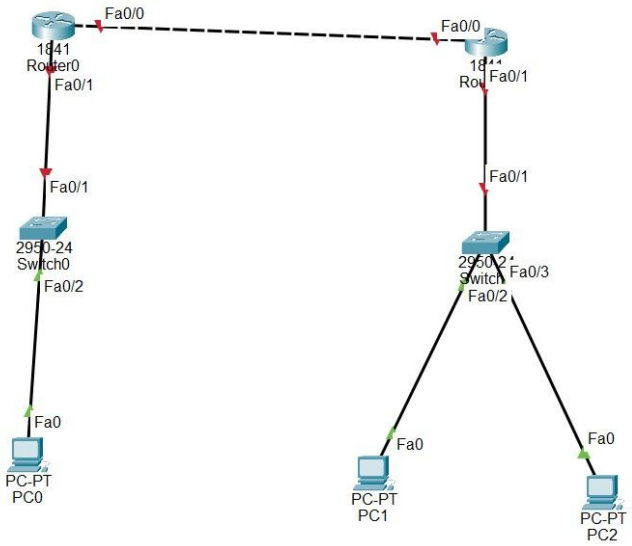


Fig 1: Network topology

A2. Configure IP Address on PCs

Click the PC → Desktop → IP Configuration → select Static and enter the values from the table.

Step 2: PC0: IP 192.168.1.2, Mask 255.255.255.0, Gateway 192.168.1.1.

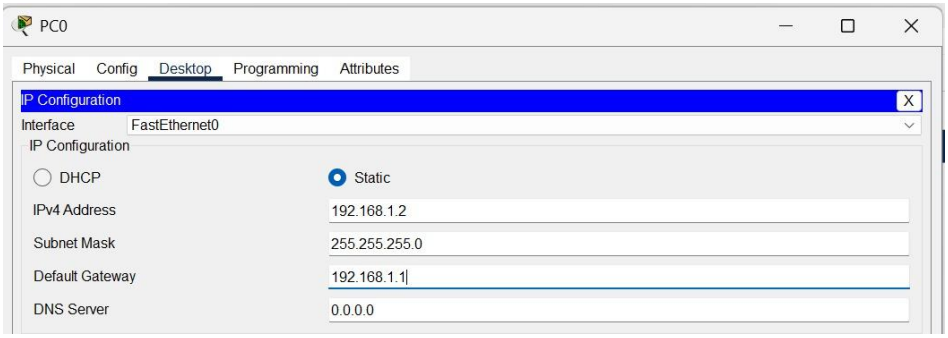


Fig 2: IP configuration of PC0

Step 3: PC1: IP 192.168.2.2, Mask 255.255.255.0, Gateway 192.168.2.1.

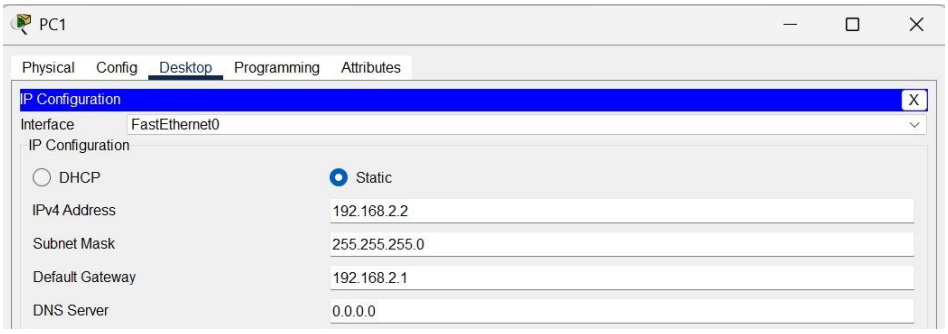


Fig 3: IP configuration of PC1

Step 4: PC2: IP 192.168.2.3, Mask 255.255.255.0, Gateway 192.168.2.1.

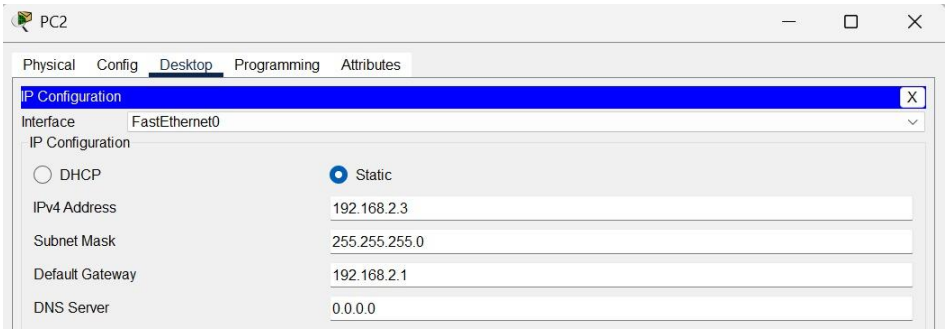


Fig 4: IP configuration of PC2

A3. Configure Router Interfaces

Click the router → Config → select the interface. Enter the IP address and subnet mask, and make sure **Port Status is ticked On** for every interface.

Step 5: Router0 – FastEthernet0/1: 192.168.1.1 / 255.255.255.0, Port Status On.

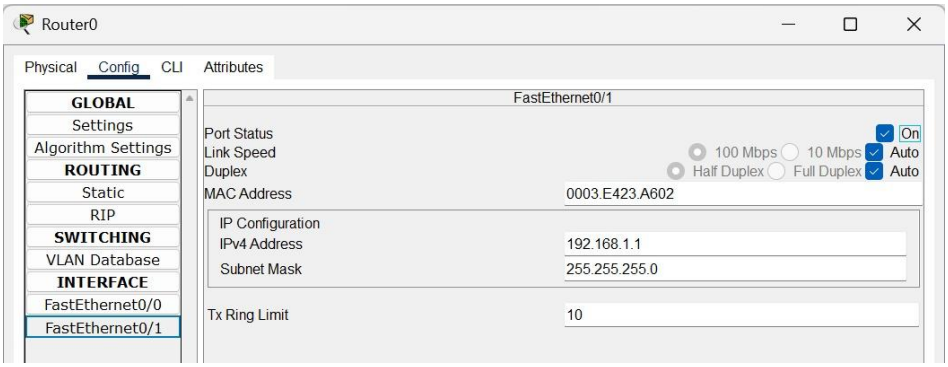


Fig 5: Router0 FastEthernet0/1

Step 6: Router0 – FastEthernet0/0: 10.0.0.1 / 255.0.0.0, Port Status On.

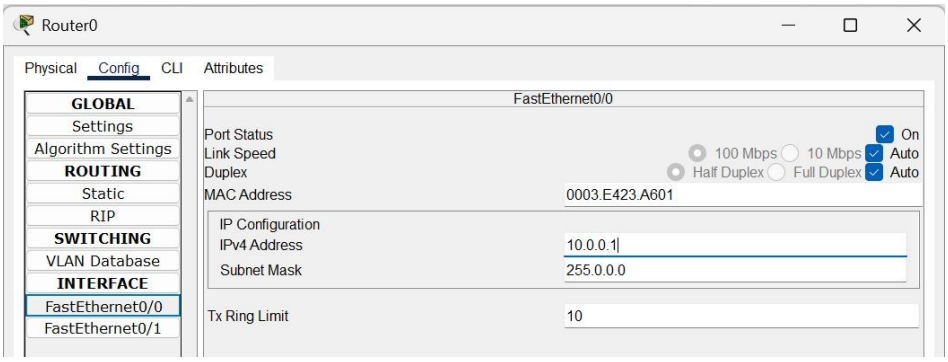


Fig 6: Router0 FastEthernet0/0

Step 7: Router1 – FastEthernet0/0: 10.0.0.2 / 255.0.0.0, Port Status On.

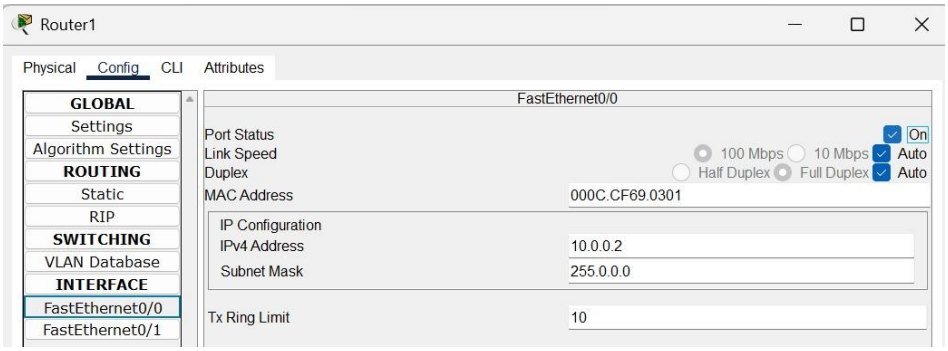


Fig 7: Router1 FastEthernet0/0

Step 8: Router1 – FastEthernet0/1: 192.168.2.1 / 255.255.255.0, Port Status On.

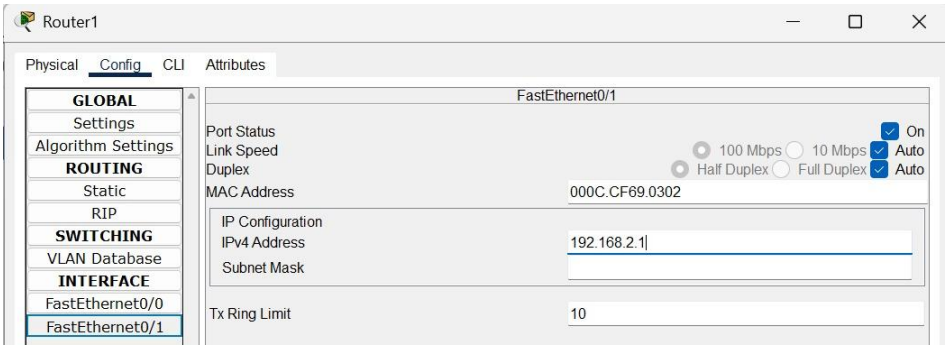


Fig 8: Router1 FastEthernet0/1

A4. Configure RIP

Click the router → Config → ROUTING → RIP. Type each directly connected network in the Network box and click Add.

Step 9: Router0: add networks 10.0.0.0 and 192.168.1.0.

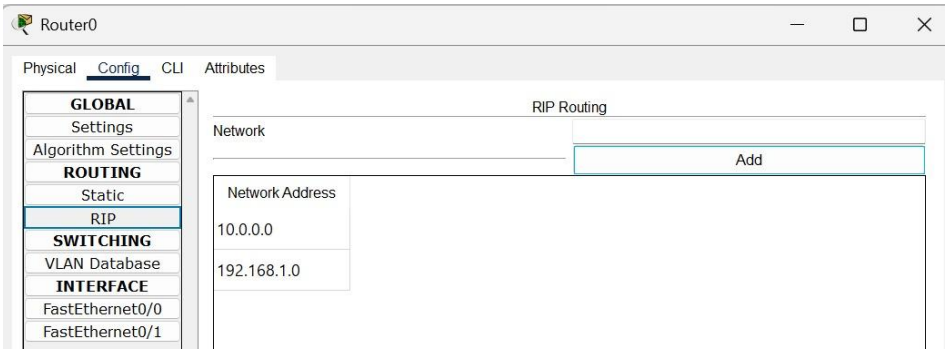


Fig 9: RIP configuration on Router0

Step 10: Router1: add networks 10.0.0.0 and 192.168.2.0.

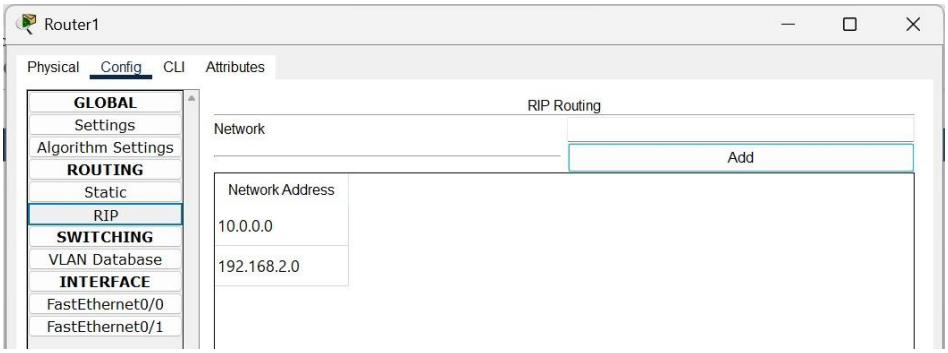


Fig 10: RIP configuration on Router1

A5. Verify RIP

Step 11: Switch from Realtime mode to Simulation mode (bottom-right corner).

Step 12: Add a Simple PDU (closed envelope icon) from PC0 to PC1.

Step 13: Click Play to start the simulation. The transmission is complete only after the ACK of the packet returns to PC0; the status in the PDU list shows Successful.

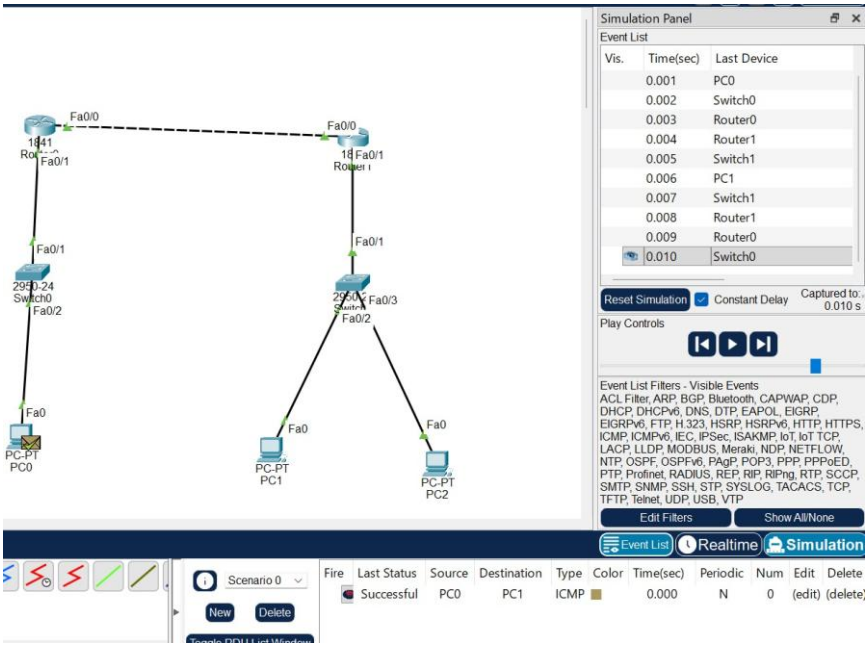


Fig 11: Simulation mode – PDU from PC0 to PC1 is Successful

Step 14: Switch back to Realtime mode. On PC0 open Desktop → Command Prompt and ping PC2 (the first request may time out because of ARP; run it again).

```
C:\> ping 192.168.2.3
```

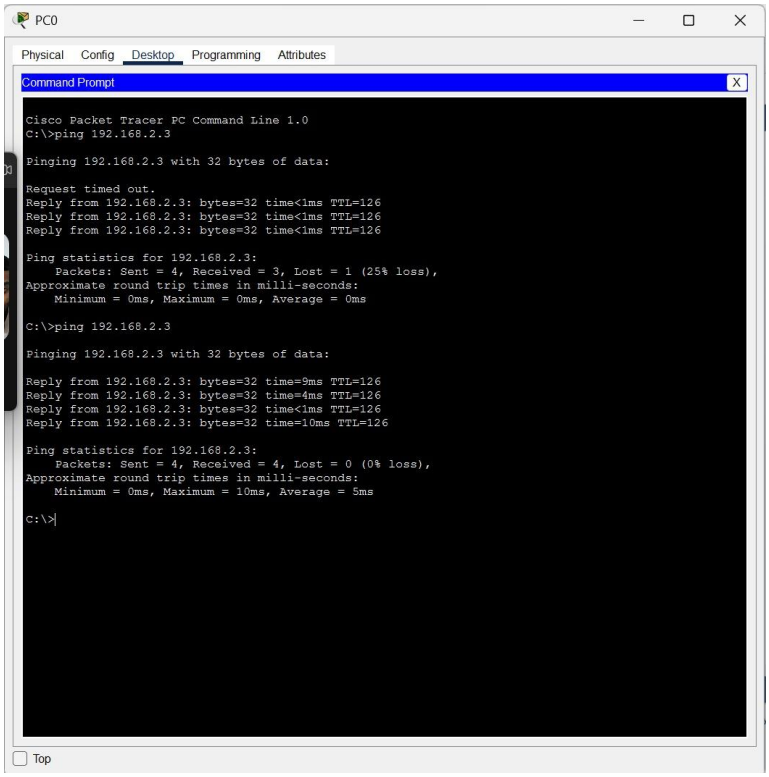


Fig 12: Ping from PC0 to PC2 – replies received (TTL=126, packet passed through 2 routers)

RIP Result: PC0 (192.168.1.0 network) can reach PC1/PC2 (192.168.2.0 network) through RIP routing.

PART B: VLAN Configuration

B1. Add Laptops

Step 15: Add 2 laptops (Laptop0 and Laptop1) and connect them to Switch1 on Fa0/4 and Fa0/5.

B2. Create VLANs on Switch1

Step 16: Open Switch1 → CLI. Press Enter and type enable (press Enter). Nothing visible happens, but it activates privileged mode. Without it, commands like config t give “Invalid input”.

Step 17: Create VLAN 20 (Purchase) and VLAN 30 (Sales) and assign ports: Fa0/2, Fa0/3 to VLAN 20 and Fa0/4, Fa0/5 to VLAN 30.

```
Switch> enable
Switch# config t
Switch(config)# vlan 20
Switch(config-vlan)# name purchase
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name sales
Switch(config-vlan)# exit
Switch(config)# int fa0/2
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
Switch(config)# int fa0/3
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
Switch(config)# int fa0/4
Switch(config-if)# switchport access vlan 30
Switch(config-if)# exit
Switch(config)# int fa0/5
Switch(config-if)# switchport access vlan 30
Switch(config-if)# exit
Switch(config)# exit
Switch#
```

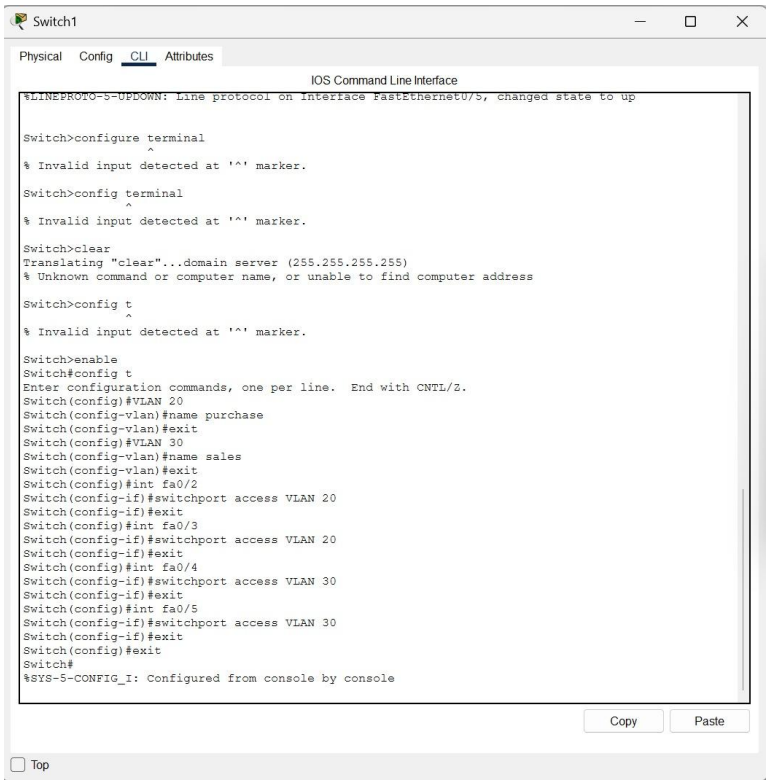


Fig 13: VLAN configuration on Switch1 CLI

B3. Configure Laptops

Step 18: Laptop0: IP 192.168.2.4, Mask 255.255.255.0, Gateway 192.168.2.1.

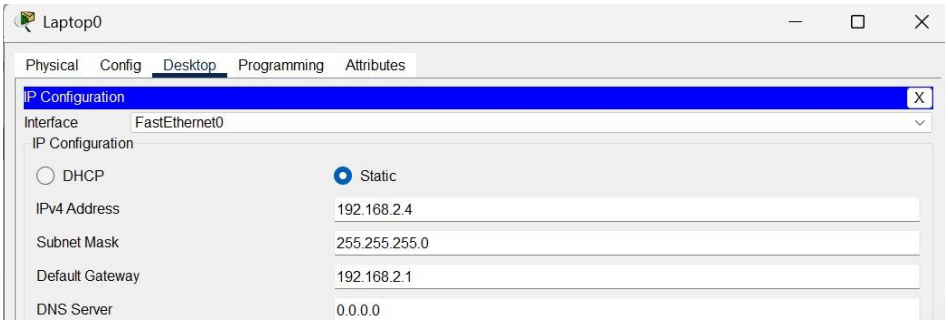


Fig 14: IP configuration of Laptop0

Step 19: Laptop1: IP 192.168.2.5, Mask 255.255.255.0, Gateway 192.168.2.1.

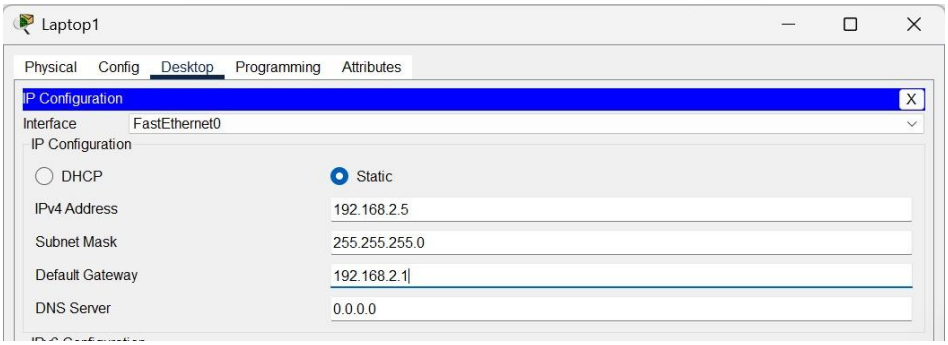


Fig 15: IP configuration of Laptop1

B4. Final Topology

Yellow = VLAN 20 (Purchase): PC1, PC2. **Pink = VLAN 30 (Sales):** Laptop0, Laptop1.

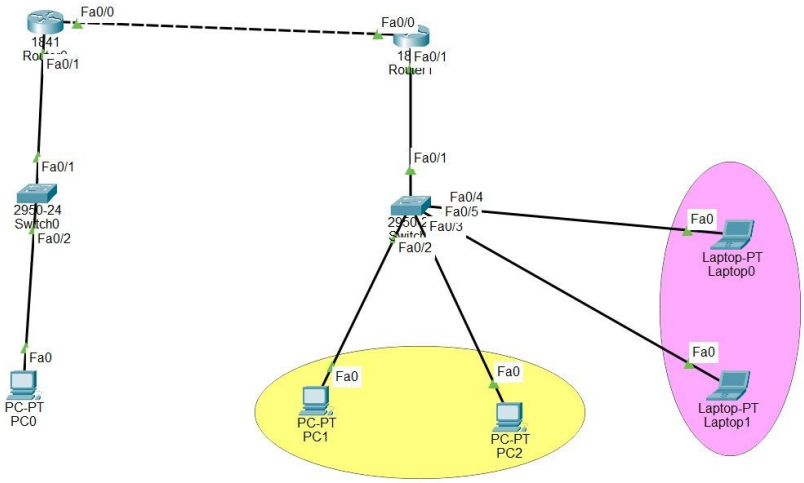


Fig 16: Final topology with VLAN 20 (yellow) and VLAN 30 (pink)

Verification

Step 20: Verify the VLANs on Switch1: type show vlan brief and check that Fa0/2–Fa0/3 are in VLAN 20 and Fa0/4–Fa0/5 are in VLAN 30.

```
Switch# show vlan brief
```

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
20   purchase                         active    Fa0/2, Fa0/3
30   sales                            active    Fa0/4, Fa0/5
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Fig 17: Output of show vlan brief – VLAN 20 (Fa0/2, Fa0/3) and VLAN 30 (Fa0/4, Fa0/5) active

Step 21: Verify by ping. From PC1 (VLAN 20) ping PC2 (same VLAN) – should be successful. Then ping Laptop0 (VLAN 30) – should fail with "Request timed out", because different VLANs cannot communicate without a router/Layer-3 device.

```
C:\> ping 192.168.2.3      (PC1 → PC2, same VLAN: Reply received)
C:\> ping 192.168.2.4      (PC1 → Laptop0, different VLAN: Request timed
out)
```

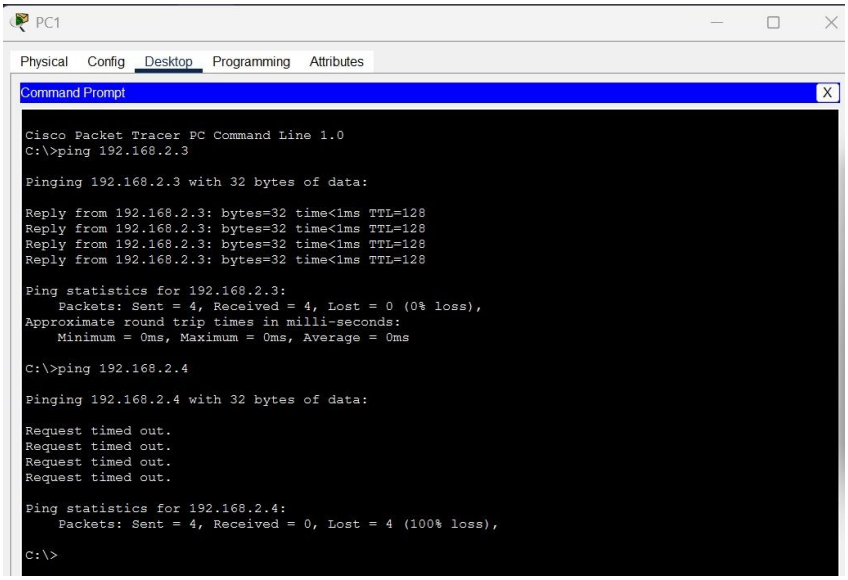


Fig 18: Ping results – same VLAN succeeds, different VLAN fails

Step 22: Also test as instructed: ping from PC0 to PC1 and from PC0 to Laptop0 and note the result. Since the router link (Switch1 Fa0/1) stays in VLAN 1 while PC1 and Laptop0 are in VLAN 20/30, these pings are not expected to reach them after VLAN creation.

Result

✓ EXPERIMENT COMPLETED SUCCESSFULLY

RIP routing was configured between Router0 and Router1 and hosts of both networks could communicate. VLAN 20 (Purchase) and VLAN 30 (Sales) were created on Switch1, and traffic was confined within each VLAN.

VIVA QUESTIONS AND ANSWERS

A. Basic Networking Concepts

Q1. What is a computer network?

Ans: A computer network is a group of devices (computers, switches, routers, printers) connected together so they can exchange data and share resources such as files, internet access and printers. In this experiment, the PCs, laptops, switches and routers together form a small network.

Q2. What is an IP address?

Ans: An IP address is a logical address given to every device on a network so that it can be identified and found. IPv4 addresses are 32 bits long, written as four decimal numbers like 192.168.1.2. It has a network part and a host part, and the subnet mask tells where the split is.

Q3. What is a subnet mask?

Ans: The subnet mask separates the network portion of an IP address from the host portion. For 255.255.255.0 the first three octets are the network and the last octet is the host, so 192.168.1.2 belongs to network 192.168.1.0. The device uses it to decide whether the destination is local or must be sent to the gateway.

Q4. What is a default gateway? Why is it needed?

Ans: The default gateway is the IP address of the router interface that a host uses to reach other networks. If PC0 wants to talk to 192.168.2.2, it sees that the address is not in its own network and hands the packet to 192.168.1.1. Without a gateway, a host can only communicate inside its own subnet.

Q5. What is a router and what does it do?

Ans: A router is a Layer-3 device that connects different networks and forwards packets between them based on the destination IP address. It looks up its routing table to choose the best path. In this experiment, Router0 and Router1 connect the 192.168.1.0 and 192.168.2.0 networks.

Q6. What is a switch?

Ans: A switch is a Layer-2 device that connects devices inside one LAN. It learns MAC addresses and forwards a frame only to the port where the destination is connected, instead of to everyone. This makes it faster and more efficient than a hub.

Q7. Difference between router and switch?

Ans: A switch works at Layer 2 using MAC addresses and connects devices within the same network, while a router works at Layer 3 using IP addresses and connects different networks. A switch cannot route between networks on its own, and a router by default separates broadcast domains on each of its interfaces.

Q8. What is a MAC address?

Ans: A MAC address is a 48-bit hardware address burned into a network interface card, written in hexadecimal such as 0003.E423.A602. It is used for delivery inside the local network (Layer 2), whereas the IP address is used for delivery across networks.

Q9. Difference between hub, switch and router?

Ans: A hub is a Layer-1 device that repeats every signal to all ports, causing collisions and wasting bandwidth. A switch sends frames only to the correct port using MAC addresses. A router connects separate networks and forwards packets using IP addresses.

Q10. Why did we use a cross-over cable between the routers?

Ans: Cross-over cables are used to connect similar devices (router-router, switch-switch, PC-PC) because the transmit pair of one side must connect to the receive pair of the other. Different devices such as router-switch or switch-PC use a straight-through cable. Packet Tracer shows the cross-over cable as a dashed line.

Q11. What is ARP?

Ans: Address Resolution Protocol finds the MAC address that belongs to a known IP address. Before sending a packet on the local network, a host broadcasts an ARP request, and the owner of that IP replies with its MAC. This is also why the first ping often fails and the following ones succeed.

Q12. What is ICMP?

Ans: Internet Control Message Protocol is a Layer-3 helper protocol used for error reporting and diagnostics. The ping command sends ICMP echo requests and receives ICMP echo replies. The Simple PDU used in simulation is also an ICMP packet.

Q13. What is a broadcast domain? What is a collision domain?

Ans: A broadcast domain is the group of devices that receive a broadcast frame sent by any one of them; a collision domain is the group of devices that share a medium and whose frames can collide. Each switch port is its own collision domain. Routers and VLANs divide a network into separate broadcast domains.

B. Routing and RIP

Q14. What is routing?

Ans: Routing is the process of finding the best path for a packet to travel from the source network to the destination network. Routers perform it by checking the destination IP against their routing table and sending the packet to the next hop.

Q15. What are the types of routing?

Ans: There are three types: static routing (manually entered routes), dynamic routing (routes learned automatically through protocols like RIP, OSPF and EIGRP) and default routing (a single route used when no specific match exists). This experiment uses dynamic routing with RIP.

Q16. Difference between static and dynamic routing?

Ans: In static routing, the administrator types every route by hand; it is simple and uses no bandwidth but does not adjust if a link fails. In dynamic routing, routers exchange information and update the table themselves, so it adapts to changes and suits larger networks, but it uses CPU and bandwidth.

Q17. What is RIP?

Ans: Routing Information Protocol is a dynamic, distance-vector routing protocol meant for small networks. Each router periodically tells its neighbours about the networks it knows, and the hop count decides the best route. It is easy to configure, which is why it is used in this practical.

Q18. What is the metric used by RIP?

Ans: RIP uses hop count, which is the number of routers a packet must pass through to reach the destination. A directly connected network has 0 hops, and each router crossed adds 1. In the screenshot, 192.168.2.0 shows [120/1], where 1 is the hop count.

Q19. What is the maximum hop count in RIP? What happens at 16?

Ans: The maximum usable hop count is 15. A route with a metric of 16 is treated as unreachable and is removed. This limit prevents endless looping but also restricts RIP to small networks.

Q20. How often does RIP send updates?

Ans: RIP sends its entire routing table to its neighbours every 30 seconds. This is visible in show ip protocols: “Sending updates every 30 seconds, next due in 11 seconds”. If no update is heard for 180 seconds, the route is marked invalid.

Q21. What is the administrative distance of RIP?

Ans: The administrative distance of RIP is 120. It is a measure of how trustworthy a route source is; when two protocols offer the same route, the lower AD wins (connected = 0, static = 1, EIGRP = 90, OSPF = 110, RIP = 120). It is the first number in [120/1] in the routing table.

Q22. Difference between RIP v1 and v2?

Ans: RIPv1 is classful: it does not send the subnet mask in updates, uses broadcast and has no authentication. RIPv2 is classless: it carries the subnet mask (supports VLSM and CIDR), uses multicast 224.0.0.9 and supports authentication. In our screenshot, the router sends v1 and receives any version by default.

Q23. What type of algorithm does RIP use?

Ans: RIP is a distance-vector protocol based on the Bellman-Ford algorithm. Routers only know the distance (hops) and direction (next hop) to networks and learn them from neighbours instead of having the full map of the network.

Q24. Which port/transport does RIP use?

Ans: RIP runs over UDP port 520. UDP is used because updates are sent periodically anyway, so there is no need for the overhead of a connection.

Q25. What is a routing table?

Ans: A routing table is the database inside a router that lists destination networks, how they were learned (connected, static, RIP), the metric, the next hop and the exit interface. The router consults it for every packet it forwards.

Q26. What is a next hop?

Ans: The next hop is the IP address of the next router on the path to the destination. In the show ip route output, 192.168.2.0 is reachable “via 10.0.0.2”, which is Router1, so 10.0.0.2 is the next hop for Router0.

Q27. Why did we add 10.0.0.0 and 192.168.1.0 in RIP on Router0?

Ans: RIP only advertises, and sends updates on, the directly connected networks that are listed with the network command. Router0 is connected to 10.0.0.0 (towards Router1) and 192.168.1.0 (towards Switch0), so both must be added so that Router1 learns about 192.168.1.0.

Q28. What would happen if we do not add a network in RIP?

Ans: That network would not be advertised, so the neighbouring router would never learn it. The packets would reach the other router but the reply or the forward path would be missing, causing ping to fail.

Q29. What if Router1 did not run RIP?

Ans: Router0 would learn nothing about 192.168.2.0, and Router1 would know nothing about 192.168.1.0. PC0 would not be able to reach PC1 or PC2 unless static routes were added on both routers.

Q30. Why is the subnet mask of 10.0.0.1 given as 255.0.0.0?

Ans: 10.0.0.0 is a Class A network and the default mask of Class A is 255.0.0.0 (/8). Both ends of the link must use the same mask, otherwise the routers will not treat the link as one network. RIPv1 also relies on the default classful mask.

Q31. Which commands are used to configure RIP in CLI?

Ans: The same RIP setup done from the GUI can be done from the CLI. In global configuration mode, type: router rip, then version 2 (optional), then network 10.0.0.0 and network 192.168.1.0. The GUI Config → RIP tab does exactly this in the background.

Q32. Which command displays the routing table?

Ans: The command is show ip route, typed in privileged mode (after enable). Directly connected networks are marked “C” and RIP-learned networks are marked “R”. See the figure below.

```
%SYS-5-CONFIG_I: Configured from console by console
enable
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
C    192.168.1.0/24 is directly connected, FastEthernet0/1
R    192.168.2.0/24 [120/1] via 10.0.0.2, 00:00:11, FastEthernet0/0

Router#
```

Fig 19: Output of show ip route on Router0

Q33. What does the line "R 192.168.2.0/24 [120/1] via 10.0.0.2, 00:00:11, FastEthernet0/0" mean?

Ans: "R" says the route was learned by RIP. 192.168.2.0/24 is the destination network. [120/1] means administrative distance 120 and metric 1 hop. "via 10.0.0.2" is the next-hop router, 00:00:11 is the time since the last update and FastEthernet0/0 is the exit interface. This line proves that RIP is working.

Q34. What do the two "C" lines in the routing table show?

Ans: "C" stands for connected. 10.0.0.0/8 on Fa0/0 and 192.168.1.0/24 on Fa0/1 are the networks directly attached to Router0, and they are added automatically when the interface has an IP address and is up.

Q35. Which command shows RIP settings and timers?

Ans: show ip protocols shows which routing protocol is running, the update timers, the version, the networks being advertised and the neighbours (routing sources). See the figure below.

```
Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 11 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version
  Interface             Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0       12 1
  FastEthernet0/1       12 1
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
        10.0.0.0
        192.168.1.0
Passive Interface(s):
Routing Information Sources:
        Gateway         Distance      Last Update
        10.0.0.2              120      00:00:24
Distance: (default is 120)
Router#
```

Fig 20: Output of show ip protocols on Router0

Q36. Explain the output of show ip protocols.

Ans: It shows that RIP is running, updates are sent every 30 seconds, a route is invalid after 180 seconds, hold-down is 180 seconds and the route is flushed after 240 seconds. It shows we send version 1 and receive any version, that automatic summarization is enabled, and that the networks being advertised are 10.0.0.0 and 192.168.1.0. The routing source 10.0.0.2 (Router1) has an AD of 120.

Q37. What is automatic summarization in RIP?

Ans: With automatic summarization (default in RIPv1 and RIPv2), a router summarizes subnets to their classful boundary when advertising across a different major network. It reduces the table size but can break discontiguous networks, so it is often disabled using no auto-summary.

Q38. What is split horizon? Route poisoning? Hold-down timer?

Ans: Split horizon prevents a router from advertising a route out of the same interface it learned it from, which avoids simple loops. Route poisoning advertises a failed route with metric 16 so everyone knows it is dead. The hold-down timer makes a router ignore new information about a failed route for a while, to avoid accepting wrong, outdated updates.

Q39. What is the count-to-infinity problem?

Ans: When a network fails, two routers can keep telling each other that they can reach it through the other, increasing the hop count each time. This continues until the metric reaches 16 (infinity for RIP). Split horizon, poison reverse and hold-down timers are used to prevent it.

Q40. What is convergence?

Ans: Convergence is the state when all routers in the network have a consistent, up-to-date view of the topology. After a link change, the faster the convergence, the lesser the packet loss. RIP converges slowly because of its 30-second updates and timers.

Q41. Why is RIP not used in large networks?

Ans: RIP is limited to 15 hops, sends the full table every 30 seconds (wasting bandwidth), converges slowly and considers only hop count (ignores bandwidth and delay). Hence, protocols like OSPF and EIGRP are used in large networks.

Q42. Name other dynamic routing protocols.

Ans: OSPF (link-state, open standard), EIGRP (Cisco advanced distance-vector), IS-IS (link-state) and BGP (path-vector, used between internet providers). RIP and EIGRP are IGPs used inside an organisation, and BGP is an EGP used between organisations.

Q43. What happens if the two ends of the router link have different subnet masks?

Ans: The two interfaces will consider themselves to be in different networks, and may not recognise each other as neighbours. RIP updates may not be accepted and the routes will fail to appear, so the mask must be the same on both ends.

C. Packet Tracer, Simulation and Ping

Q44. Why do we use Simulation mode?

Ans: Simulation mode slows down the network so we can watch every packet move from device to device, along with the protocol (ARP, ICMP) and the OSI-layer details at each hop. It helps us understand how data actually travels and where a failure happens.

Q45. What is a Simple PDU?

Ans: A Simple PDU (the closed envelope tool) is a quick test packet, essentially a single ICMP echo request, which Packet Tracer sends between two chosen devices. If the reply comes back, the status in the PDU list shows “Successful”.

Q46. Difference between Realtime and Simulation mode?

Ans: Realtime mode runs the network continuously at normal speed, like a live network. Simulation mode lets us step through events one at a time with an event list, so it is used for study and troubleshooting rather than normal operation.

Q47. When is the transmission complete in simulation?

Ans: It is complete when the ACK (echo reply) returns to the source, PC0. Only then does the PDU status change to “Successful”, because success means a full round trip, not just delivery to the destination.

Q48. What is the use of the ping command?

Ans: Ping checks reachability and response time to another device. It sends ICMP echo requests and waits for echo replies, and reports packets sent/received/lost and round-trip time. It is the first test to run after configuring IPs and routing.

Q49. Why did the first ping show “Request timed out”?

Ans: At the first ping, the devices do not know each other’s MAC addresses, so ARP has to be performed at each hop, which takes longer than the ping timeout. The first packet is lost, but the ARP tables are filled, so the rest succeed (as in our 25% loss on the first attempt and 0% on the second).

Q50. What does TTL=126 mean?

Ans: TTL (Time To Live) is reduced by 1 at every router. A Windows PC starts with 128, and the packet from PC0 to PC2 passed through two routers (Router0 and Router1), so 128 − 2 = 126. It proves the packet crossed two routers.

Q51. Why was TTL=128 when PC1 pinged PC2?

Ans: PC1 and PC2 are connected through the same switch, in the same network and VLAN. No router was crossed, so TTL was not decreased and stayed at the original value of 128.

Q52. What is the meaning of "Request timed out"?

Ans: It means no reply was received within the time limit. The causes could be a missing route, wrong IP/gateway, a link that is down, a firewall, ARP delay or, as in our experiment, the destination being in a different VLAN.

Q53. What does "Destination host unreachable" mean?

Ans: It is an ICMP message from a router saying it has no route to the destination network. It differs from "Request timed out", where nobody replied at all.

Q54. Why must Port Status be On?

Ans: Port Status On means the interface is administratively enabled. If it is off (shutdown), the port will not send or receive anything and the link stays down, so the router will not even place the connected network in its routing table.

Q55. What do the green and red triangles on links mean?

Ans: Green triangles show that the link is up and working. Red triangles show the link is down, which commonly happens in Packet Tracer when the interface is off or the link has not completed negotiating yet. In our topology all links are green before testing.

Q56. What is the difference between Static and DHCP in IP configuration?

Ans: With Static, we manually type the IP, mask and gateway, which gives a fixed predictable address. With DHCP, a server automatically assigns these settings. We used Static here so the addressing plan is exactly as designed.

D. VLAN Concepts

Q57. What is a VLAN?

Ans: A Virtual LAN is a logical division of a switch into separate networks, so that devices are grouped by function instead of physical location. Each VLAN is its own broadcast domain, as if each had its own separate switch.

Q58. Why are VLANs used?

Ans: VLANs are used to reduce broadcast traffic, increase security by isolating departments, make management easier and improve performance. Here, the Purchase and Sales departments are isolated even though they connect to the same switch.

Q59. What are the advantages of VLANs?

Ans: Smaller broadcast domains improve performance, departments are isolated for security, users can be moved logically without re-cabling, fewer switches are needed (cost saving) and the network is easier to manage and troubleshoot.

Q60. What is the default VLAN? Which VLAN do all ports belong to by default?

Ans: VLAN 1 is the default VLAN and all ports belong to it until changed. In our show vlan brief, all unused ports (Fa0/1, Fa0/6 to Fa0/24) are still in VLAN 1. VLAN 1 cannot be deleted or renamed.

Q61. Why can devices in different VLANs not communicate?

Ans: Each VLAN is a separate broadcast domain, so ARP broadcasts do not cross to another VLAN, and the switch will not forward frames between them. To communicate, traffic must be routed by a Layer-3 device. This is why PC1 could not ping Laptop0.

Q62. How do we make different VLANs communicate?

Ans: Through inter-VLAN routing: either router-on-a-stick (one trunk link to a router with sub-interfaces, one per VLAN, each acting as the gateway) or a Layer-3 switch using SVIs (Switch Virtual Interfaces).

Q63. What are VLAN 20 and VLAN 30 in this experiment?

Ans: VLAN 20 is named “purchase” and contains the yellow group: PC1 and PC2 (ports Fa0/2 and Fa0/3). VLAN 30 is named “sales” and contains the pink group: Laptop0 and Laptop1 (ports Fa0/4 and Fa0/5).

Q64. What is an access port?

Ans: An access port belongs to a single VLAN and carries untagged traffic. It is used to connect end devices such as PCs and laptops. The command switchport access vlan 20 makes the port an access port in VLAN 20.

Q65. What is a trunk port?

Ans: A trunk port carries traffic of many VLANs over one link, between switches or between a switch and a router. The frames are tagged with the VLAN ID (802.1Q) so the other side knows which VLAN they belong to.

Q66. What is 802.1Q?

Ans: IEEE 802.1Q is the standard for VLAN tagging on trunk links. It inserts a 4-byte tag into the Ethernet frame containing the 12-bit VLAN ID. The frames of the native VLAN are sent without a tag.

Q67. What is the range of normal VLAN IDs?

Ans: Normal-range VLANs are 1 to 1005 and extended-range are 1006 to 4094 (12-bit ID = 4096 values). VLANs 1 and 1002–1005 are default ones (fddi, token-ring, etc., as in the show vlan brief output) and cannot be deleted.

Q68. What command creates a VLAN?

Ans: In global configuration mode, type vlan 20 and then name purchase. This creates VLAN 20 and gives it a name; the name is optional but makes it easy to identify.

Q69. What command assigns a port to a VLAN?

Ans: Enter the interface with int fa0/2, then type switchport access vlan 20. The port is now a member of VLAN 20 and is removed from VLAN 1 automatically.

Q70. What command is used to verify VLANs?

Ans: show vlan brief lists every VLAN, its name, status and member ports. This is the main confirmation that the VLAN part is done correctly. See the figure below.

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
20   purchase                         active    Fa0/2, Fa0/3
30   sales                            active    Fa0/4, Fa0/5
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Fig 21: Output of show vlan brief on Switch1

Q71. How do you read the show vlan brief output?

Ans: VLAN 1 (default) lists the unused ports Fa0/1 and Fa0/6 to Fa0/24. VLAN 20 “purchase” is active with ports Fa0/2 and Fa0/3, and VLAN 30 “sales” is active with ports Fa0/4 and Fa0/5. The VLANs 1002–1005 are default system VLANs. This matches the configuration, so the VLAN setup is correct.

Q72. Why did we type “enable” first?

Ans: The switch starts in user EXEC mode (Switch>) with limited commands. enable moves to privileged EXEC mode (Switch#), where we can use config t and show commands. It prints nothing, but the prompt changes from > to #.

Q73. What does “config t” mean?

Ans: It is short for configure terminal. It moves from privileged mode into global configuration mode (Switch(config)#), where settings such as VLANs and interfaces can be changed.

Q74. Why did we get “Invalid input detected at ^ marker”?

Ans: Because the command was typed in user mode (Switch>), where configure terminal is not available. The ^ points to the place where the switch stopped understanding the command. After typing enable, the same command works.

Q75. Why did PC1 ping PC2 successfully but fail to ping Laptop0?

Ans: PC1 and PC2 are in the same VLAN (20) so the switch forwards frames between them directly. Laptop0 is in VLAN 30, a different broadcast domain, so ARP and ping cannot reach it and the result is “Request timed out”, even though all IPs are in 192.168.2.0/24.

Q76. What if all ports are put in the same VLAN?

Ans: Then all devices are in one broadcast domain and communicate freely, exactly as with no VLANs. The isolation is lost, so VLANs only make sense when ports are in different VLANs.

Q77. What happens if a port is not assigned to any VLAN?

Ans: It stays in the default VLAN 1. All such ports can talk to each other, which is not secure, so unused ports are usually put in an unused VLAN and shut down.

Q78. What happens if we assign a port to a VLAN that does not exist?

Ans: On Cisco switches, the VLAN is created automatically, in most versions, as soon as the command is applied (and a message may be shown). Hence we create the VLAN first and name it properly before assigning ports, as is done in the experiment.

Q79. What is the native VLAN?

Ans: The native VLAN is the VLAN whose traffic is sent untagged on a trunk, and by default it is VLAN 1. Both ends of a trunk must have the same native VLAN, or a mismatch warning and leaks between VLANs can happen.

Q80. The laptops are in the same subnet as PC1. Why do they not communicate?

Ans: The IP subnet is a Layer-3 concept, and VLAN separation works at Layer 2. A host first needs the destination’s MAC via an ARP broadcast, and the broadcast is limited to its own VLAN. Since the ARP never reaches the other VLAN, the ping fails despite the same IP network.

Q81. What is VTP?

Ans: VLAN Trunking Protocol is a Cisco protocol that automatically shares VLAN names and IDs from one switch (server) to the other switches in the same domain, so VLANs need not be created on each switch manually. It is not needed in this single-switch experiment.

Q82. How many broadcast domains are there on Switch1 after creating VLANs?

Ans: Three: VLAN 1 (router port Fa0/1 and unused ports), VLAN 20 and VLAN 30. Before creating VLANs, the whole switch was one broadcast domain.

Q83. Where is VLAN information stored?

Ans: In a file called vlan.dat in the switch’s flash memory, not in the startup-config. Hence it survives a reboot even if the config is not saved, and it is deleted separately using delete flash:vlan.dat.

Q84. What is the difference between a LAN and a VLAN?

Ans: A LAN is a physical network defined by the devices and cables connected together, while a VLAN is a logical network defined in the switch software. A single physical LAN can be divided into many VLANs without changing the cabling.

Q85. Types of VLANs?

Ans: Data VLAN (normal user traffic), Voice VLAN (IP phones), Management VLAN (to manage switches), Native VLAN (untagged trunk traffic) and Default VLAN (VLAN 1). Our VLAN 20 and 30 are data VLANs.

Q86. What is the difference between static and dynamic VLAN?

Ans: In a static VLAN, the administrator assigns each port to a VLAN manually (what we did). In a dynamic VLAN, a server (VMPS) assigns the VLAN automatically based on the MAC address of the connecting device.

Q87. Which layer do VLANs operate on?

Ans: VLANs work at Layer 2 (Data Link layer) because they separate frames and broadcast domains inside the switch. Making them talk to each other requires Layer 3 (routing).

Q88. How is a subnet related to a VLAN?

Ans: Normally one VLAN is mapped to one IP subnet, for example VLAN 20 to 192.168.20.0/24. This keeps the design clean. In our experiment, both VLANs share 192.168.2.0/24, which is why the pings fail even though they look like the same network; in a real design they should use different subnets.

Q89. What is inter-VLAN routing and why was it not done here?

Ans: Inter-VLAN routing enables communication between different VLANs using a router or Layer-3 switch. It was not configured because the purpose of this practical is to show VLAN isolation. It can be added later with a trunk port and router sub-interfaces.

E. Scenario / "What if" Questions

Q90. What if the cable between Router0 and Router1 is removed?

Ans: The link goes down and Router0 removes the directly connected 10.0.0.0 network. The RIP routes learned through it become invalid after 180 seconds and are flushed after 240 seconds, so PC0 can no longer reach PC1 or PC2.

Q91. What if Router0 Fa0/0 and Router1 Fa0/0 are in different networks?

Ans: They will not be able to exchange RIP updates, as RIP only works between neighbours in a shared subnet. The routers would not learn each other's LANs and the ping across them would fail.

Q92. What if the default gateway on PC0 is wrong?

Ans: PC0 will be able to talk only to devices in 192.168.1.0/24. For any other network it will send the packet to a wrong or non-existent gateway, so ping to PC1/PC2 will time out even if RIP is perfect.

Q93. What if the same IP is given to two devices?

Ans: An IP address conflict occurs: both devices get confused replies and connectivity becomes intermittent. Each device must have a unique IP address in a network.

Q94. What if the PC subnet mask is wrong?

Ans: The PC calculates the local network incorrectly, so it may try to reach a remote host directly using ARP (and fail), or send a local host's packet to the gateway. This leads to ping failures even though the IPs look fine.

Q95. What if the router interface is shut down?

Ans: The interface goes down, the connected network is removed from the routing table and RIP stops advertising it. All the hosts behind that interface lose connectivity.

Q96. What if we add a new network on Router1 but forget to add it in RIP?

Ans: Router1 will have it as connected, but will not advertise it. Router0 will never learn it, so devices behind Router0 cannot reach that new network.

Q97. What if the RIP versions on the routers are different?

Ans: If one router sends v1 and the other expects v2 only, updates can be ignored. RIPv1 does not carry masks, so routes may be wrongly interpreted. By default Cisco routers send v1 and receive both, so it still works as in our output.

Q98. What if a laptop is moved from VLAN 30 to VLAN 20?

Ans: The laptop would join the Purchase group: it can ping PC1 and PC2, and it can no longer ping Laptop1. Only the switchport access vlan command on its port needs to change; no re-cabling is needed.

Q99. What if Fa0/1 of Switch1 (router link) is put in VLAN 20?

Ans: The router would become part of VLAN 20, so PC1 and PC2 could use it as their gateway and reach Router0 and PC0, but Laptop0 and Laptop1 (VLAN 30) would have no path to the router.

Q100. What if the switch is connected to the router with a trunk port?

Ans: All VLANs can reach the router over that one link. With sub-interfaces (for example Fa0/1.20 and Fa0/1.30), the router becomes the gateway for each VLAN and enables inter-VLAN routing (router-on-a-stick).

Q101. What if we ping PC0 from Laptop0 after VLAN creation?

Ans: The ping will time out. Laptop0 is in VLAN 30, but the router port is in VLAN 1, so Laptop0's traffic can never reach its gateway 192.168.2.1.

Q102. Why does the first ping from PC0 to PC1 take longer than later pings?

Ans: The first packet needs ARP at every hop to find MAC addresses, and the routers must find the next hop. Once the ARP tables are filled, later pings go straight through, so they are faster (our second ping had 0% loss).

Q103. How can we check the routes learned through RIP?

Ans: Use show ip route (look for lines starting with R) and show ip protocols (to see the networks, timers and neighbours). Also, show ip rip database can display the RIP database in some IOS versions.

Q104. How do you see the packet details in Packet Tracer?

Ans: In Simulation mode, click on an event in the event list (or on the envelope on the topology) to open PDU Information. It shows the contents of the packet at each OSI layer as it passes through the device.

Q105. How would you save the configuration on a switch/router?

Ans: Use copy running-config startup-config (or write memory) in privileged mode. Without saving, the configuration, which is in RAM, is lost when the device is restarted.

Q106. What is the difference between running-config and startup-config?

Ans: running-config is the active configuration stored in RAM, and every command we type changes it immediately. startup-config is stored in NVRAM and is loaded when the device boots. They are the same only after we save.